# Lecture-29
# External Flows II: Laminar BL for flow over a flat plate

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# Learning Objectives

By the end of this lecture, you should be able to

1. Interpret the **Blasius similarity solution** to map the changing velocity within the laminar boundary layer.
2. Quantify the **boundary layer thickness** and **wall shear stress** from the Blasius solution

# Governing equations

- The governing equations for **mass and momentum conservation** can be simplified by applying **scaling analysis**.

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=\nu\frac{\partial^2 u}{\partial y^2}$$

$$\text{at } y=0, \quad u=0, \quad v=0$$

$$\text{at } y=\infty, \quad u=U, \quad \frac{\partial u}{\partial y}=0$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

# Blasius Self Similarity idea

- Blasius reasoned that the dimensionless velocity profile in the x-direction should be **similar** for all values of x when plotted **versus a dimensionless distance from the wall**.

$$\frac{u}{U} = g(\eta) \quad \text{where} \quad \eta \propto \frac{y}{\delta}$$

- Blasius also reasoned that the **thickness of this boundary layer** should vary as the **square root of x**.

$$\eta = y\sqrt{\frac{U}{\nu x}}$$

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# Resulting simplifications

- The **x-component of the velocity** can be simplified as

$$u = \frac{\partial \psi}{\partial y} = \frac{\partial \psi}{\partial \eta}\frac{\partial \eta}{\partial y} = \sqrt{\nu x U}\,\frac{df}{d\eta}\sqrt{\frac{U}{\nu x}} = U\frac{df}{d\eta}$$

- The **y-component of the velocity** can be simplified as

$$v = \frac{1}{2}\sqrt{\frac{\nu U}{x}}\left[\eta\frac{df}{d\eta} - f\right]$$

- The **momentum equation** gets simplified to

$$2\frac{d^3 f}{d\eta^3} + f\frac{d^2 f}{d\eta^2} = 0$$

$$\text{at } \eta = 0, \quad f = \frac{df}{d\eta} = 0$$

$$\text{at } \eta = \infty, \quad \frac{df}{d\eta} = 1$$

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# Solutions of the governing equation

- For a value of **η equal to 5**, the velocity becomes **approximately 99% of the free-stream velocity**. Recall that we defined the boundary layer thickness using this idea!!
- Boundary layer thickness

$$\delta \approx \frac{5.0}{\sqrt{U/\nu x}} = \frac{5.0x}{\sqrt{Re_x}}$$

- Wall shear stress

$$\tau_w = 0.332U\sqrt{\rho\mu U/x} = \frac{0.332\rho U^2}{\sqrt{Re_x}}$$

| $\eta = y\sqrt{\frac{U}{\nu x}}$ | $f$ | $f' = \frac{u}{U}$ | $f''$ |
|---|---|---|---|
| 0 | 0 | 0 | 0.3321 |
| 0.5 | 0.0415 | 0.1659 | 0.3309 |
| 1.0 | 0.1656 | 0.3298 | 0.3230 |
| 1.5 | 0.3701 | 0.4868 | 0.3026 |
| 2.0 | 0.6500 | 0.6298 | 0.2668 |
| 2.5 | 0.9963 | 0.7513 | 0.2174 |
| 3.0 | 1.3968 | 0.8460 | 0.1614 |
| 3.5 | 1.8377 | 0.9130 | 0.1078 |
| 4.0 | 2.3057 | 0.9555 | 0.0642 |
| 4.5 | 2.7901 | 0.9795 | 0.0340 |
| 5.0 | 3.2833 | 0.9915 | 0.0159 |
| 5.5 | 3.7806 | 0.9969 | 0.0066 |
| 6.0 | 4.2796 | 0.9990 | 0.0024 |
| 6.5 | 4.7793 | 0.9997 | 0.0008 |
| 7.0 | 5.2792 | 0.9999 | 0.0002 |
| 7.5 | 5.7792 | 1.0000 | 0.0001 |
| 8.0 | 6.2792 | 1.0000 | 0.0000 |

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# Revisiting the learning objectives

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